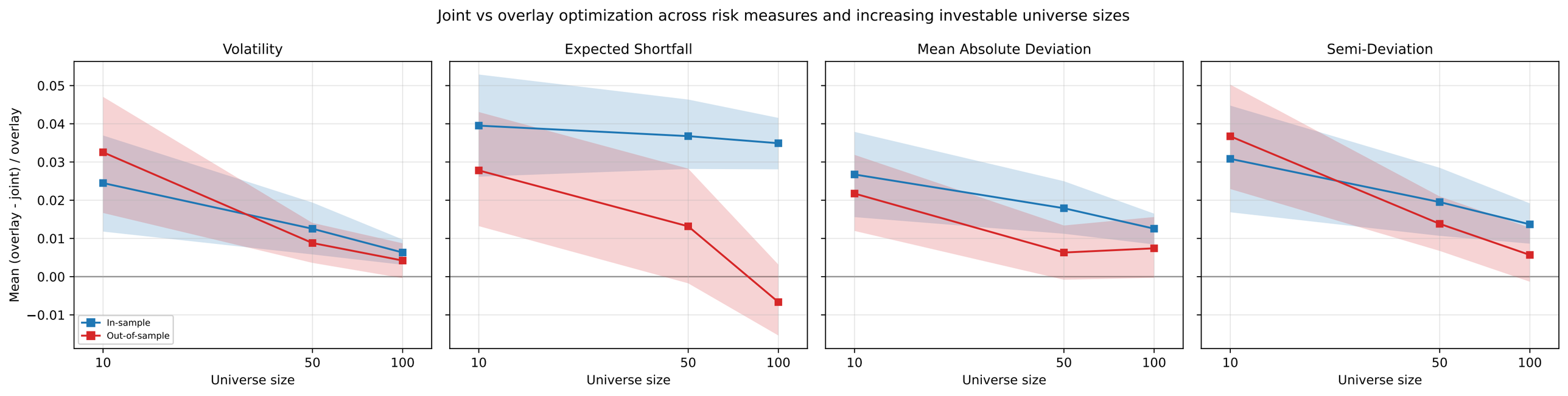
Joint vs overlay optimization across risk measures and increasing investable universe sizes
Volatility
Expected Shortfall
Mean Absolute Deviation
Semi-Deviation
Mean (overlay - joint) / overlay
0.05
0.04
0.03
0.02
0.01
0.00
−0.01
In-sample
Out-of-sample
10
50
100
Universe size